

Math 300 Exam 2

Key

Show all your work in the space provided for each problem. All problems will be graded for correctness, completeness, conciseness, and clarity. Do your best. Don't cut corners.

1. (18 pts) Refer to the Cayley table to answer the questions about the group G .

a) (2 pts) What is the identity in the group?

1

b) (2 pts) What is $5 \cdot 8$?

2

c) (3 pts) Prove that G is not Abelian.

*	1	2	3	4	5	6	7	8
1	1	2	3	4	5	6	7	8
2	2	3	4	1	8	5	6	7
3	3	4	1	2	7	8	5	6
4	4	1	2	3	6	7	8	5
5	5	6	7	8	3	4	1	2
6	6	7	8	5	2	3	4	1
7	7	8	5	6	1	2	3	4
8	8	5	6	7	4	1	2	3

d) (2 pts) What is the order of the group?

8

e) (3 pts) What is the order of the element 5?

4

f) (2 pts) What is the centralizer of 3 in G ?

All of G

$\{1, 2, 3, 4, 5, 6, 7, 8\}$

g) (2 pts) What is the centralizer of 8 in G ?

$\{1, 3, 6, 8\}$

h) (2 pts) What is 5^{-1} in G ?

7

2. (3 pts) Give examples of three groups that have 6 elements. (One bonus point if you can give a fourth group with 6 elements)

$$\mathbb{Z}_6, S_3, D_3, U(7), U(9), U(14)$$

3. (8 pts)

a) (4 pts) Use the Euclidean algorithm to find the $\gcd(5723, 2328)$

$$5723 = 2 \cdot 2328 + 1067$$

$$2328 = 2 \cdot 1067 + 194$$

$$1067 = 5 \cdot 194 + \boxed{97}$$

$$194 = 2 \cdot 97 + 0$$

$$\boxed{\gcd(5723, 2328) = 97}$$

b) (4 pts) Find integers x and y such that $\gcd(5723, 2328) = 5723x + 2328y$.

$$97 = 1067 - 5 \cdot 194$$

$$= 1067 - 5(2328 - 2 \cdot 1067) = 11 \cdot 1067 - 5 \cdot 2328$$

$$= 11 \cdot (5723 - 2 \cdot 2328) - 5 \cdot 2328$$

$$= 11 \cdot 5723 - 27 \cdot 2328$$

$$\boxed{x = 11 \quad y = -27}$$

4. (13 pts) Let S be the set of all 4-digit numbers from 1000 to 9999. If x and y are elements of S we can define a relation $\sim$ on S by $x \sim y$ if and only if their first digits (digits on the far left) are the same.

a) (2 pts) Describe two different equivalence classes for this relation.

$$[1000] = \{x \in S : 1000 \leq x < 2000\}$$

$$[2000] = \{x \in S : 2000 \leq x < 3000\}$$

b) (2 pts) How many different equivalence classes are there for this relation.

There are 9 classes, one for each of the 9 possible first digits.

c) (9 pts) Prove that $\sim$ is an equivalence relation on the set S .

$\sim$ is reflexive: For all $x \in S$ $x \sim x$ since the 1st digit of x is the same as the first digit of x .

$\sim$ is symmetric: [show if $x \sim y$ then $y \sim x$].

Let x and y be in S and let $x \sim y$.

This means the first digit of x and y are the same.

Therefore the first digit of y and x are the same.

So $y \sim x$.

$\sim$ is transitive: [show if $x \sim y$ and $y \sim z$, then $x \sim z$]

Let x, y, z be in S and let $x \sim y$ and $y \sim z$.

This means the first digit of x equals the first digit of y and the first digit of y equals the first digit of z .

Therefore the first digit of x equals the first digit of z .

So $x \sim z$.

$\sim$ is an equivalence relation on S since it is reflexive, symmetric, and transitive.

5. (4 pts) Explain in precise mathematical language and notation what each of the following statements means.

a) The group G is Abelian means ...

$$\text{For all } x \text{ and } y \text{ in } G \quad x \cdot y = y \cdot x.$$

b) The distributive law holds in a field F means ...

$$\text{For all } x, y, \text{ and } z \text{ in } F \quad x \cdot (y + z) = (x \cdot y) + (x \cdot z)$$

6. (6 pts) List or describe all elements and the binary operation which makes each set a group.

Group	Elements	Operation
$\mathbb{Z}_8$	0, 1, 2, 3, 4, 5, 6, 7	Addition mod 8
$\mathbb{Q}^*$	Non zero rational numbers	Multiplication

7. (6 pts) Give examples of two Pythagorean triples of the form $[9, b, c]$ where $9 < b < c$.

$$[9, 12, 15] \quad \text{using formula } [3k, 4k, 5k] \text{ with } k=3$$

$$[9, 40, 41] \quad \text{using formula } \left[n, \frac{n^2-1}{2}, \frac{n^2+1}{2}\right] \text{ with } n=9$$

8. (6 pts) Prove there are infinitely many Pythagorean triples.

Use either of the formulas in #7 to generate infinitely many Pythagorean triples.

We know $[3, 4, 5]$ is a P.T. since $3^2 + 4^2 = 5^2$.

Let k be any positive integer. Then $[3k, 4k, 5k]$ is a P.T.

Since all three of these are positive integers and

$$(3k)^2 + (4k)^2 = 9k^2 + 16k^2 = 25k^2 = (5k)^2$$

9. (10 pts) Prove that if p and q are distinct primes then $\sqrt{pq}$ is irrational.

Suppose for contradiction that $\sqrt{pq}$ is rational.

This means $\sqrt{pq} = \frac{a}{b}$ with $a, b \in \mathbb{Z}$, $b \neq 0$, $\gcd(a, b) = 1$

This leads to $pgb^2 = a^2$. Now $p \mid pgb^2$ since $gb^2 \in \mathbb{Z}$.

Therefore $p \mid a^2$ and by Euclid's Lemma $p \mid a$.

This means $a = pk$ for some $k \in \mathbb{Z}$.

Substituting into the equation $pgb^2 = a^2$ gives us $pgb^2 = p^2k^2$ which simplifies to $gb^2 = pk^2$.

$p \mid pk^2$ since $k^2 \in \mathbb{Z}$. Therefore $p \mid gb^2$. But $p \nmid g$ since p and g are distinct primes. Thus $p \mid b^2$ by Euclid's Lemma and we get $p \mid b$. But $p \mid a$ and $p \mid b$ contradicts $\gcd(a, b) = 1$ so $\sqrt{pq}$ must be irrational.

10. (6 pts) If p and q are distinct primes prove that $\sqrt{p+q}$ could be rational or irrational.

If $p=2$ and $q=3$ $\sqrt{p+q} = \sqrt{5}$ is irrational

If $p=2$ and $q=7$ $\sqrt{p+q} = \sqrt{9} = 3$ is rational

11. Permutation problems (10 pts)

(2 pts) What is the order of the symmetric group S_7 ?

$$7! = 5040$$

(2 pts) What is the order of the element $\alpha = (153)(2647)$?

$$\text{lcm}(3, 4) = 12$$

(4 pts) Write the following product of permutations in standard form.

$$(26)(1273)(2354)(1325)(2653) = (1\ 5)(3\ 4\ 7)$$

(2 pts) If $\beta = (123456)$, what is β^{20} ?

$$\beta^{20} = (\beta^6)^3 \cdot \beta^2 = \beta^2 = (1\ 3\ 5)(2\ 4\ 6)$$

12. (5 pts) Find integers a and b such that $2.34\overline{56} = \frac{a}{b}$

$$\text{Let } x = 2.34\overline{56} \quad \text{so } 100x = 234.\overline{56}$$

$$10000x = 23456.\overline{56}$$

$$9900x = 10000x - 100x = 23456.\overline{56} - 234.\overline{56} = 23222$$

$$\text{So } x = \frac{23222}{9900}$$

$$a = 23222 \quad b = 9900$$

13. (5 pts) Find all positive integer solutions to the Diophantine equation $xy + 7x - 5y = 0$.

Rearranging the equation we get $7x = 5y - xy = y(5-x)$

For both x and y to be positive we must have $0 < x < 5$

x	1	2	3	4
$y = \frac{7x}{5-x}$	$\frac{7}{4}$	$\frac{14}{3}$	$\frac{21}{2}$	28

The only positive integer solution is

$$x = 4, y = 28$$